

Numerical PDEs Homework #0

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Problem 1: Finite differences

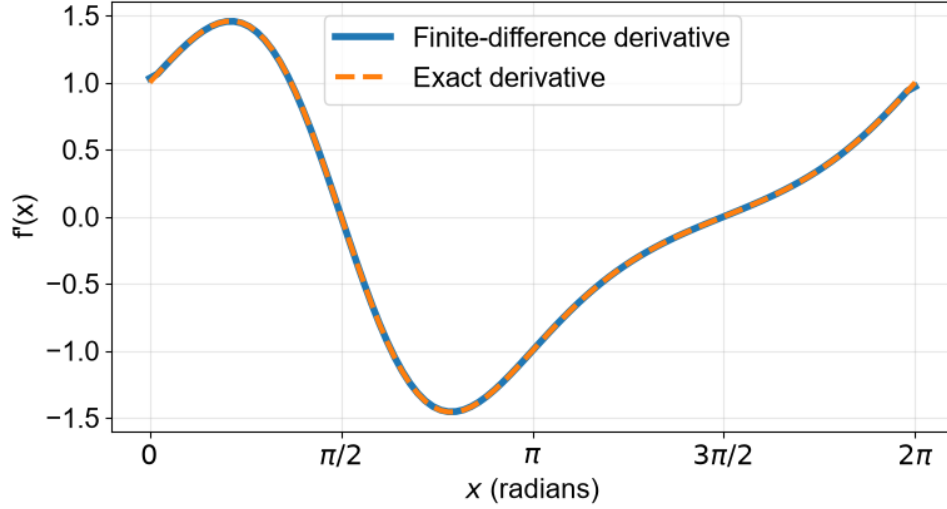


Figure 1: Finite difference approximation for $f(x) = \exp(\sin(x))$.

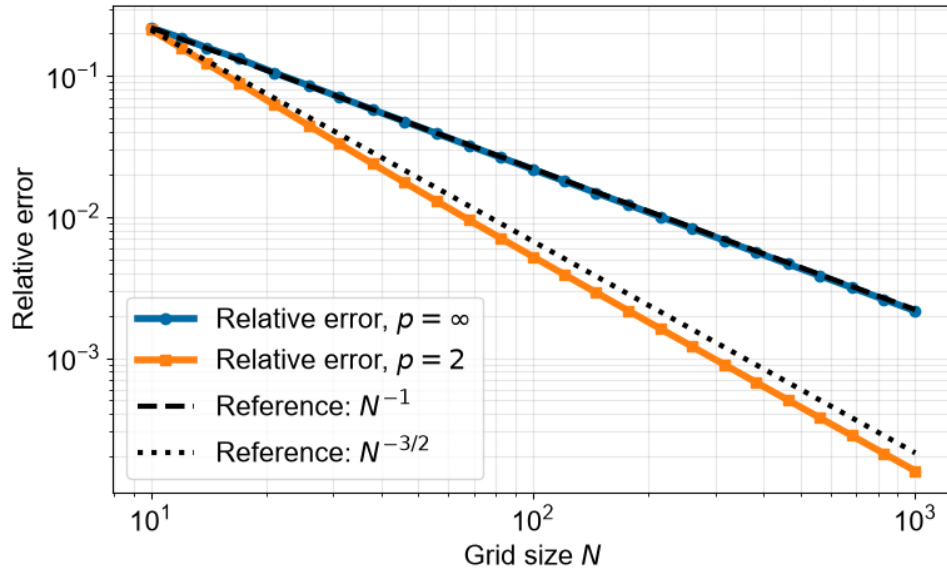


Figure 2: Relative-error convergence of first derivative finite differences for $f(x) = \exp(\sin(x))$

I implemented this solution using the centered difference equation for first derivatives on interior points with forward and backward differences on the endpoints. As seen in Figure 1, the numerical approximation is visually indistinguishable from the exact solution, but differs slightly at the endpoints due to the first-order endpoint differences having higher error ($\mathcal{O}(h)$)

vs second-order $\mathcal{O}(h^2)$).

Figure 2 shows the difference in relative error slope for $p = 2$ and $p = \infty$ norms. The $p = \infty$ scales as the largest error in the approximation, which will be the $\mathcal{O}(h)$ error at the endpoints, giving the slope of 1. The $p = 2$ norm will be weighted more evenly, where the interior $\mathcal{O}(h^2)$ errors will effect the relative error, giving the $3/2$ slope.

Problem 2: Second-order derivatives

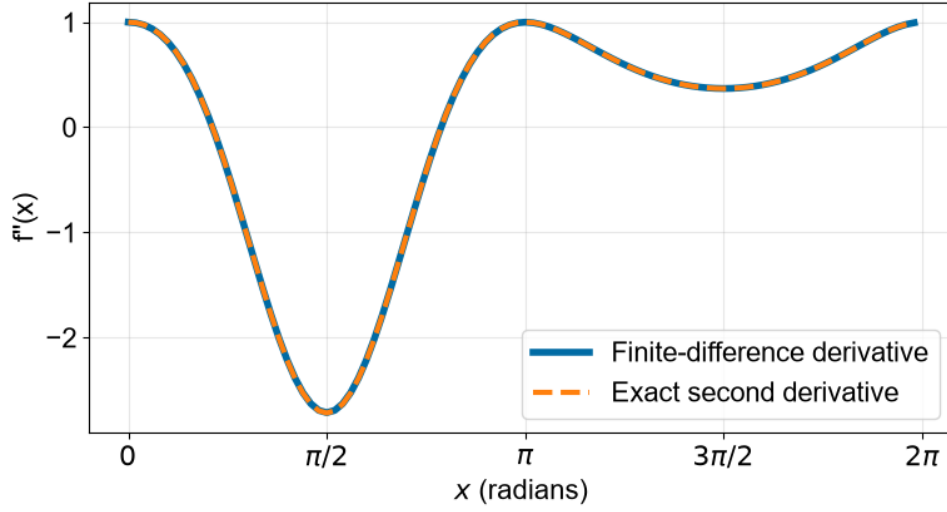


Figure 3: Numerical second derivative of $f(x) = \exp(\sin(x))$

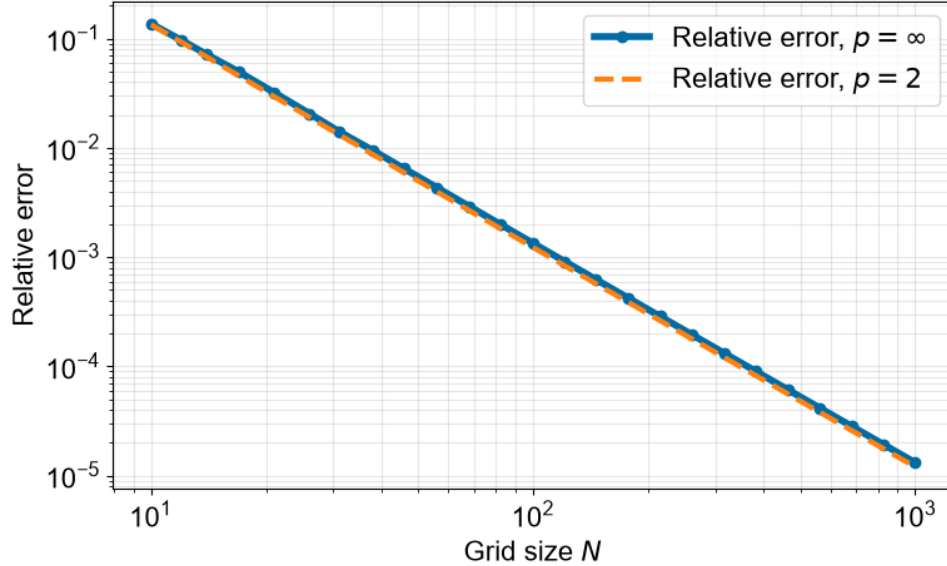


Figure 4: Relative-error convergence of second derivative finite difference for $f(x) = \exp(\sin(x))$.

I implemented this solution using the same method as part one with periodic assumptions instead of first-order differences at the endpoints. Figure 3 shows that the numerical approximation is visually indistinguishable from the exact solution on the interior grid points again, and also on the endpoints because the periodic assumption reduced the error at those points. Figure 4 shows that the relative-error converges at the same rate for both norms, which is expected since the error is $\mathcal{O}(h^2)$ at all points.

Problem 3: ODE

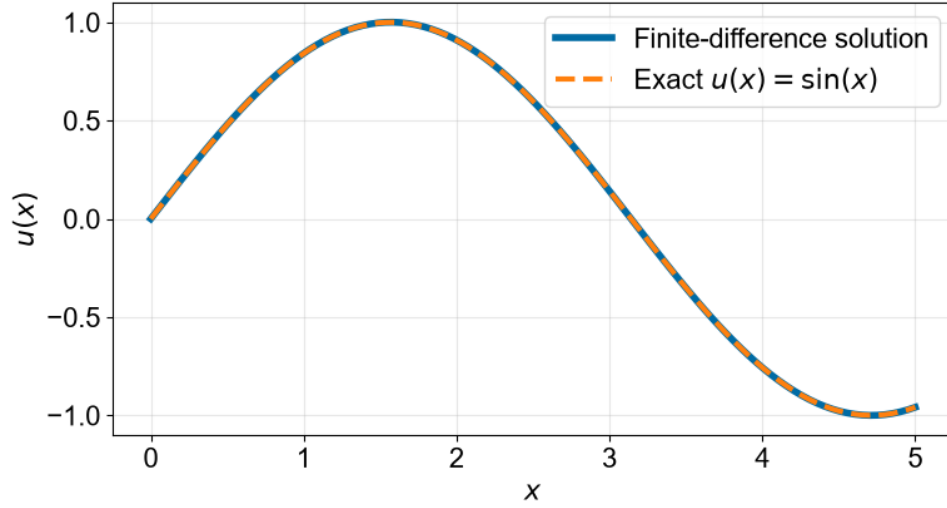


Figure 5: Finite-difference solution of $\frac{d^2u}{dx^2} + \sin(x)\frac{du}{dx} + u(x) = f(x)$, with $u(x) = \sin(x)$

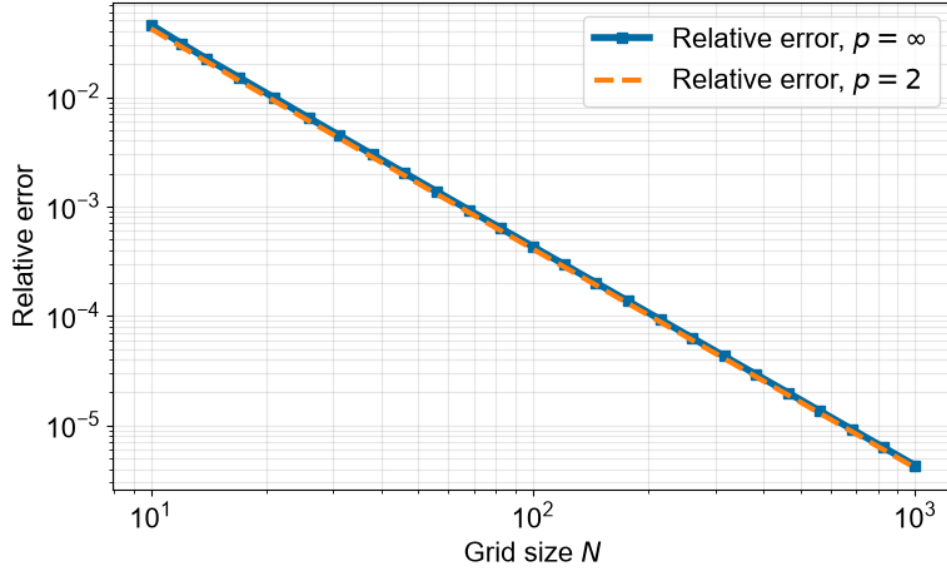


Figure 6: Relative-error convergence of ODE solution

I implemented this solution by constructing a sparse matrix and setting the boundary conditions on the right hand side. I used `scipy` for the sparse matrix and linear system solver. Again, Figure 5 shows that the numerical approximation is visually indistinguishable from the exact solution, and Figure 6 shows the relative-error converges at the same rate for both norms.

Problem 4: Poisson Equation

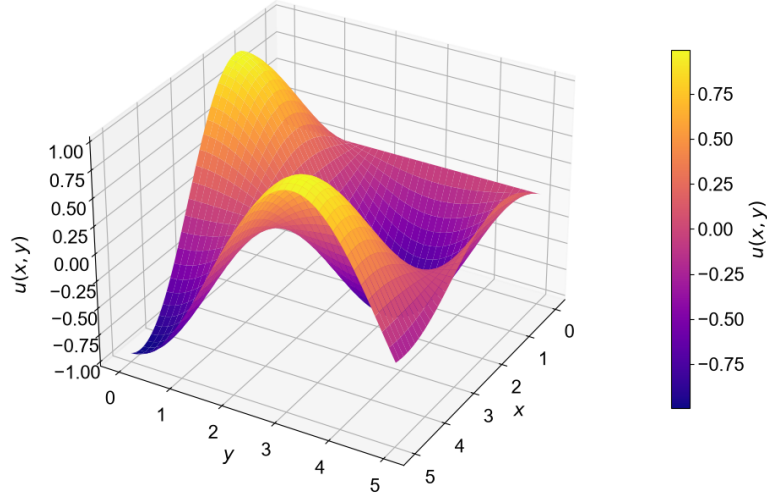


Figure 7: 3D finite-difference solution of $\nabla^2 u = f(x, y)$

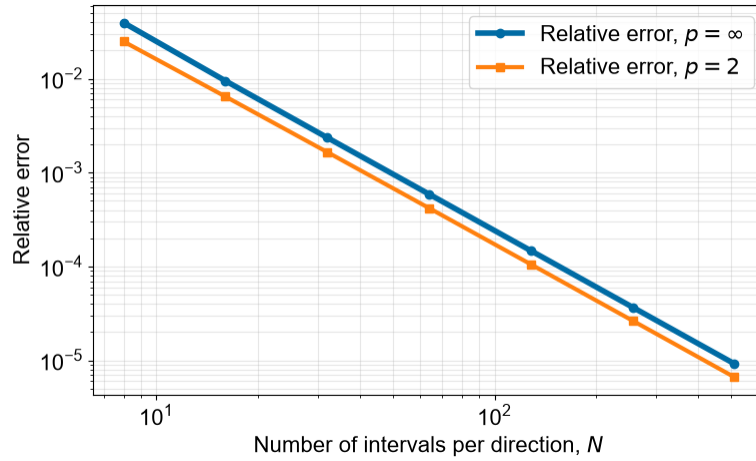


Figure 8: Relative-error convergence of the 2D Poisson solver

I implemented this solution by constructing a sparse matrix from kronecker products and flattening the unknowns and right hand side matrices. I set the boundary conditions in the right hand side using a boundary index mapping. The approximate surface is shown in Figure 7, and Figure 8 shows that the relative error decreases similarly to the previous problems, thus giving an accurate approximation. The difference in the relative-error for $p = 2$ and $p = \infty$ is explained by different norm constants, but they still converge at the same rate.

Neumann boundary conditions do not work because they do not give a unique solution. Since they only specify derivatives, for any solution u , $u + C$ is also a solution. In linear algebra, the Neumann conditions gives a laplacian that is not invertible and therefore cannot

satisfy the linear system with a unique solution. A solution to this problem could be setting a single condition for $u(x, y)$ to effectively fix a single unique solution in place.